

Generalized linear models

Estimation in logit models

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Estimation in logit models - Contents

Estimation in logit models

LS estimation

$$\underline{x} = (1, x_1, \dots, x_p)^T$$

$$E[y | \underline{x}] = P(y = 1 | \mathbf{x}) = \frac{\exp(\mathbf{x}^T \boldsymbol{\beta})}{1 + \exp(\mathbf{x}^T \boldsymbol{\beta})}$$

How could $\boldsymbol{\beta}$ be estimated by means of the least squares method?

$$\sum_{i=1}^n \left(y_i - \underbrace{\frac{\exp(\underline{x}_i^T \boldsymbol{\beta})}{1 + \exp(\underline{x}_i^T \boldsymbol{\beta})}}_{\pi(\underline{x}_i)} \right)^2 \rightarrow \min$$

LS estimation

Problem

Corresponds to the following assumptions:

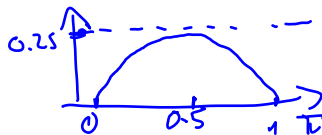
$$\pi(x_i) = \mathbb{E}(y | \mathbf{x}_i) = \frac{\exp(\mathbf{x}_i^\top \beta)}{1 + \exp(\mathbf{x}_i^\top \beta)}$$



$$\text{Var}(y | \mathbf{x}_i) = \text{const.}$$



as $\text{Var}(y | x_i) = \pi(x_i)(1 - \pi(x_i))$



ML estimation

$$\vec{x}_i = (1, x_{i1}, \dots, x_{ip})^T$$

- ▶ Given the data $(y_i, \mathbf{x}_i)$, $i = 1, \dots, n$
- ▶ $y_i \in \{0, 1\}$, independent given $\mathbf{x}_i$
- ▶ $\mathbf{x}_i$ already properly coded

Likelihood function

$$\begin{aligned} L(\beta) &= f(y_1, \dots, y_n \mid \beta, \mathbf{x}_1, \dots, \mathbf{x}_n) \\ &= \prod_{i=1}^n f(y_i \mid \beta, \mathbf{x}_i) \\ &= \prod_{i=1}^n \pi(\mathbf{x}_i)^{y_i} (1 - \pi(\mathbf{x}_i))^{1-y_i} \end{aligned}$$

ML estimation

Log-likelihood

$$\begin{aligned}l(\beta) = \log L(\beta) &= \sum_{i=1}^n (y_i \log \pi(\mathbf{x}_i) + (1 - y_i) \log(1 - \pi(\mathbf{x}_i))) \\&= \sum_{i=1}^n \left(y_i \log \left(\frac{\pi(\mathbf{x}_i)}{1 - \pi(\mathbf{x}_i)} \right) + \log(1 - \pi(\mathbf{x}_i)) \right) \\&\quad \text{logits} \hat{=} \mathbf{x}_i^T \beta \quad \quad \quad \hat{=} 1 - \frac{\exp(\mathbf{x}_i^T \beta)}{1 + \exp(\mathbf{x}_i^T \beta)}\end{aligned}$$

$$\boxed{= \sum_{i=1}^n (y_i \mathbf{x}_i^T \beta - \log(1 + \exp(\mathbf{x}_i^T \beta)))} = \frac{1}{1 + \exp(\mathbf{x}_i^T \beta)}$$

Next steps?

differentiable w.r.t. β , and set to zero

Recap: Derivatives w.r.t. vectors

Case 1: $\mathbb{R}^p \rightarrow \mathbb{R}$ $(x_1, \dots, x_p) \mapsto f(\underline{x})$

$$\frac{\partial f}{\partial \underline{x}} = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_p} \right)^T$$

Case 2: $\mathbb{R}^p \rightarrow \mathbb{R}^q$ $(x_1, \dots, x_p) \mapsto f_1(\underline{x}), \dots, f_q(\underline{x})$

$$\Rightarrow \frac{\partial f}{\partial \underline{x}} = \left(\frac{\partial f_1}{\partial \underline{x}}, \dots, \frac{\partial f_q}{\partial \underline{x}} \right) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_p} \\ \vdots & & \vdots \\ \frac{\partial f_q}{\partial x_1} & \dots & \frac{\partial f_q}{\partial x_p} \end{pmatrix}$$

ML estimation

Score function

$$s(\beta) = \frac{\partial l(\beta)}{\partial \beta} = \left(\frac{\partial l(\beta)}{\partial \beta_1}, \dots, \frac{\partial l(\beta)}{\partial \beta_p} \right)^T$$

j-th entry

$$\text{with } \frac{\partial l(\beta)}{\partial \beta_j} = \frac{\partial}{\partial \beta_j} \left(\sum_{i=1}^n y_i \underbrace{x_i^T \beta}_{\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}} - \log(1 + \exp(x_i^T \beta)) \right)$$

$$= \sum_{i=1}^n \left(y_i x_{ij} - \frac{1}{1 + \exp(x_i^T \beta)} \cdot \exp(x_i^T \beta) \cdot x_{ij} \right)$$

$$= \sum_{i=1}^n \left(y_i - \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right) \cdot x_{ij}$$

$$= \sum_{i=1}^n (y_i - \tau(x_i)) \cdot x_{ij}$$

ML estimation

Finding the roots of a function

Equation system:

$$s(\beta) = 0$$

$$\Leftrightarrow \sum_{i=1}^n (y_i - \pi(\underline{x}_i)) \underline{x}_i = \underline{0}$$

$$\Rightarrow \beta = \dots \quad \text{↯}$$

- ▶ No analytical closed form solution as in LS
- ▶ Only iteratively solvable!
- ▶ One possibility: via Fisher scoring

↔ Newton Raphson